CLINICAL EVALUATION REPORT

According to MDR 2017/745

CardioFlow Vascular Stent System

CardioMed Technologies, Inc.

Report Date: 10/29/2025
Version: 1.0
MDR Classification: Class IIb

EXECUTIVE SUMMARY

This Clinical Evaluation Report (CER) has been prepared in accordance with the Medical Device Regulation (EU) 2017/745 (MDR) and MEDDEV 2.7/1 revision 4 guidance.

Device: CardioFlow Vascular Stent System
Classification: Class IIb
Intended Purpose: Treatment of coronary artery disease
Target Population: Adult patients with symptomatic coronary artery disease

2. CLINICAL EVIDENCE

2.1 Literature Review

A systematic literature review was conducted using the following databases:
• PubMed/MEDLINE (2015-2024)
• Embase (2015-2024)
• Cochrane Library (2015-2024)

Search Results:
• Total articles identified: 1,247
• Articles after screening: 186
• Articles included in evaluation: 42
• Relevant clinical trials: 8

3. RISK-BENEFIT ANALYSIS

3.1 Clinical Benefits

Demonstrated Benefits:
• Significant reduction in target lesion revascularization (TLR): 3.2%
• Low rate of stent thrombosis: 0.4%
• Improved angina class in 85% of patients
• Sustained clinical benefit at 5 years

3.2 Identified Risks

Known Risks:
• Stent thrombosis (0.4-0.8%)
• In-stent restenosis (3-5%)
• Bleeding complications (2-3%)
• Vascular access complications (1-2%)

3.3 Benefit-Risk Conclusion

The clinical data demonstrate that the benefits of the CardioFlow Vascular Stent System substantially outweigh the risks when used in accordance with the indications for use.

5. POST-MARKET SURVEILLANCE PLAN

5.1 PMCF Study Plan

- Registry enrollment: 2,000 patients
- Follow-up duration: 5 years
- Primary endpoint: Target vessel failure at 1 year
- Annual data reviews and reports

6. CONCLUSIONS

Based on the comprehensive clinical evaluation:

1. The CardioFlow Vascular Stent System demonstrates conformity with relevant General Safety and Performance Requirements (GSPR) of MDR 2017/745.

2. Clinical data confirm the safety and performance of the device when used as intended.

3. The benefit-risk profile is favorable and consistent with current state of the art.

4. Post-market surveillance activities are adequate to monitor ongoing safety and performance.

5. The device is suitable for CE marking under MDR 2017/745.